

Research Summary

Evaluating Cross-Layer Host–Network Telemetry Correlation in a Reproducible SOC Testbed

Overview

This research investigates whether correlating host-level telemetry (e.g., Sysmon and Wazuh agent logs) with network-level telemetry (Zeek network logs) can improve intrusion detection effectiveness in modern Security Operations Centers (SOCs). Contemporary SOC environments often rely on isolated detection sources, which contributes to alert fatigue, elevated false-positive rates, and increased detection latency.

To address this challenge, the project designs and deploys a reproducible, VM-based Mini SOC testbed that integrates host and network telemetry within an isolated environment. The testbed enables controlled attack execution and systematic comparison of detection performance with and without cross-layer correlation.

Research Objectives

The primary objectives of this research are:

- To evaluate the impact of cross-layer host–network telemetry correlation on intrusion detection accuracy.
- To determine whether correlated detection can reduce alert noise while improving true positive rates and detection latency compared to single-source detection approaches.
- To demonstrate that a reproducible SOC testbed can be used to empirically measure and validate these improvements across representative attack scenarios.

Methodology

The study is conducted using a four-node SOC architecture comprising an attacker host, a

Windows victim system, a Zeek-based network sensor, and a centralized SIEM platform built on the modern Wazuh stack (Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard).

Controlled attack scenarios—including authentication brute-force attempts, reconnaissance activities, and network-based data exfiltration—are executed within the isolated environment. Host and network telemetry are collected simultaneously and analyzed under two conditions: independent single-source detection and cross-layer correlated detection, enabling direct baseline comparison.

Evaluation Metrics

Detection performance is evaluated using quantitative metrics, including:

- True Positive Rate (TPR) and False Positive Rate (FPR)
- Alert volume reduction, as a measure of alert noise
- Detection latency, defined as the time from event occurrence to alert generation
- Correlation gain, quantifying the effectiveness of consolidating signals across telemetry sources

Expected Contribution

This work provides an empirical evaluation of cross-layer telemetry correlation as a practical approach to improving SOC detection capabilities. The resulting testbed, experimental methodology, and findings aim to contribute to research in SOC operations, security data analytics, and intrusion detection system evaluation, while emphasizing reproducibility and practical applicability.

Ethical Considerations

All experiments are conducted strictly within controlled, locally owned virtual machines. No external systems or networks are targeted, and all datasets released publicly are sanitized to remove identifying information. Ethical usage guidelines and experiment constraints are documented in the accompanying ETHICS.md file.